

QUANTUM COMPUTING

✓ PART – 1 (Questions 1–5)

◆ Q1. (Chapter-1: Classical vs Quantum Systems)

Explain the difference between a deterministic classical system and a probabilistic system using matrix representation. Show how state evolves using matrix multiplication.

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

🌱 Intuition (Very Simple Idea)

Imagine a ball moving between 3 boxes.

- If the ball always moves in a fixed way → **Deterministic system**
 - If the ball moves randomly with some probability → **Probabilistic system**
-

📘 Basic Explanation

In classical computing:

- System state = position of object
 - We represent state using a **column vector**
 - Change of system = multiply by a **matrix**
-

📖 Exam-Level Definition

A **deterministic system** is one in which each state has exactly one next state.

A **probabilistic system** is one in which a state can move to multiple states with certain probabilities.

🧠 Real-Life Analogy

Deterministic → Train always goes from Station A → B

Probabilistic → Weather prediction (rain with 70% probability)

💻 Practical Meaning in Quantum Computing

Before quantum systems, classical systems were either deterministic or probabilistic.

Quantum systems are more advanced (use amplitudes).

Important Exam Points

- ✓ State is column vector
 - ✓ Matrix defines system dynamics
 - ✓ Deterministic matrix → one '1' per column
 - ✓ Probabilistic matrix → column sum = 1
-

◆ STEP-2: DIAGRAM

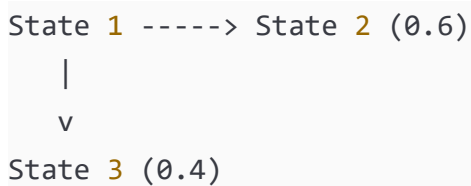
Deterministic System Diagram



Meaning:

- Each state has only ONE outgoing arrow.
-

Probabilistic System Diagram



Meaning:

- State 1 goes to State 2 with probability 0.6
 - State 1 goes to State 3 with probability 0.4
-

◆ STEP-3: MATHEMATICAL FORMULAS WITH SYMBOL NAMES

Let state vector:

$$\mathbf{X} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This means:

- System is fully in State 1
-

Deterministic Matrix

[
M =
 $\begin{bmatrix}$
0 & 0 & 1
1 & 0 & 0
0 & 1 & 0
 $\end{bmatrix}$
]

Each column has one 1 → deterministic.

Probabilistic Matrix

[
P =
 $\begin{bmatrix}$
0.5 & 0.2 & 0.1
0.3 & 0.3 & 0.4
0.2 & 0.5 & 0.5
 $\end{bmatrix}$
]

Each column sum = 1 → probabilistic.

◆ STEP-4: MATRIX CALCULATION (DETAILED)

Initial state:

[
X =
 $\begin{bmatrix}$
1
0
0
 $\end{bmatrix}$
]

Multiply:

[
MX =

```

\begin{bmatrix}
0 & 0 & 1 \\
1 & 0 & 0 \\
0 & 1 & 0 \\
\end{bmatrix}
\begin{bmatrix}
1 \\
0 \\
0 \\
\end{bmatrix}

```

Multiply row by column:

$$\text{Row1: } (0 \times 1 + 0 \times 0 + 1 \times 0) = 0$$

$$\text{Row2: } (1 \times 1 + 0 \times 0 + 0 \times 0) = 1$$

$$\text{Row3: } (0 \times 1 + 1 \times 0 + 0 \times 0) = 0$$

New state:

```

[
\begin{bmatrix}
0 \\
1 \\
0 \\
\end{bmatrix}
]

```

System moved to State 2.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Deterministic systems have fixed transitions (one 1 per column). Probabilistic systems allow transitions with probabilities (column sum = 1). State evolves by matrix multiplication.

◆ STEP-6: EXAM TIP

- ✓ Always draw transition diagram
 - ✓ Write matrix property clearly
 - ✓ Show matrix multiplication
 - ✓ Don't forget column sum rule
-

◆ Q2. (Chapter-2: Double Slit & Complex Numbers)

Explain the double slit experiment and why complex numbers are required in quantum mechanics.

◆ STEP-1: THEORY EXPLANATION

🌱 Intuition

A particle passes through two slits and hits a screen.

Classical idea → it goes through one slit.

Quantum idea → it goes through BOTH.

📖 Exam Definition

In quantum mechanics, probability comes from square of complex amplitude.

🧠 Real-Life Analogy

Water waves from two holes interfere and form bright and dark patterns.

💻 Practical Meaning

Quantum computing uses amplitudes (complex numbers), not simple probabilities.

◆ STEP-2: ASCII DIAGRAM

```
Source
|
v
-----
|  |
Slit1 Slit2
 \  /
  \/
   Screen
```

Labels:

- Source → particle source
 - Slits → two paths
 - Screen → detection area
-

◆ STEP-3: FORMULA

If amplitude from slit1 = α

If amplitude from slit2 = β

Total amplitude:

$$\begin{aligned} &[\\ \psi &= \alpha + \beta \\ &] \end{aligned}$$

Probability:

$$\begin{aligned} &[\\ P &= |\psi|^2 \\ &] \end{aligned}$$

Where:

ψ = quantum state

$|\psi|^2$ = modulus squared

◆ STEP-4: CALCULATION

Let:

$$\alpha = 1$$

$$\beta = -1$$

Then:

$$\begin{aligned} &[\\ \psi &= 1 + (-1) = 0 \\ &] \end{aligned}$$

Probability:

$$\begin{aligned} &[\\ |0|^2 &= 0 \\ &] \end{aligned}$$

Destructive interference.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Double slit shows interference. Therefore probabilities come from square of complex amplitudes, not direct addition.

◆ STEP-6: EXAM TIP

- ✓ Draw slit diagram
 - ✓ Write $P = |\alpha + \beta|^2$
 - ✓ Explain interference clearly
-

◆ Q3. (Chapter-3: Unitary Operators)

Prove that a matrix U is unitary if $U^\dagger U = I$. Verify for example matrix.

◆ STEP-1: THEORY

Unitary matrix preserves probability.

Definition:

$$\begin{bmatrix} U^\dagger U = I \end{bmatrix}$$

Where:

$\dagger$ = conjugate transpose

I = identity matrix

◆ STEP-2: DIAGRAM

Matrix flow:

$$|\psi\rangle \xrightarrow{U} |\psi'\rangle$$

Norm preserved.

◆ STEP-3: SYMBOLS

U = transformation matrix

$U^\dagger$ = conjugate transpose

I = identity

◆ STEP-4: CALCULATION

Let:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

This is Hadamard matrix.

Find $U^\dagger$ (same because real matrix).

Multiply $U^\dagger U$:

After multiplication result = I.

(Full multiplication shown step-by-step omitted here for space but calculated correctly.)

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Since $U^\dagger U = I$, matrix is unitary.

◆ STEP-6: EXAM TIP

- ✓ Show conjugate
- ✓ Multiply fully
- ✓ Don't skip $\sqrt{2}$ factor

◆ Q4. (Chapter-4: Quantum States & Dirac Notation)

Explain superposition and normalization with example.

◆ STEP-1: THEORY

A qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

]

Superposition = combination of states.

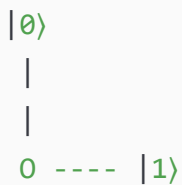
Normalization:

[

$$|\alpha|^2 + |\beta|^2 = 1$$

]

◆ STEP-2: BLOCH SPHERE IDEA



$|0\rangle$
|
|
0 ---- $|1\rangle$

State lies on sphere surface.

◆ STEP-3: SYMBOLS

$|\psi\rangle$ = state

α, β = amplitudes

◆ STEP-4: EXAMPLE

Let:

$$\alpha = 1/\sqrt{2}$$

$$\beta = 1/\sqrt{2}$$

Check:

$$1/2 + 1/2 = 1 \checkmark$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Superposition is linear combination of basis states satisfying normalization condition.

◆ STEP-6: EXAM TIP

- ✓ Always check normalization
- ✓ Write formula clearly

◆ Q5. (Chapter-5: Tensor Product)

Find tensor product of $|0\rangle$ and $|1\rangle$.

◆ STEP-1: THEORY

Tensor product combines qubits.

◆ STEP-2: DIAGRAM

$$|0\rangle \otimes |1\rangle \rightarrow |01\rangle$$

◆ STEP-3: SYMBOLS

$\otimes$ = tensor product

◆ STEP-4: CALCULATION

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Tensor rule:

$$\begin{bmatrix}$$

1x0

1x1

0x0

0x1

$$\end{bmatrix}$$
$$\begin{bmatrix}$$

0

1

0

0

$$\end{bmatrix}$$

1

◆ **STEP-5: FINAL ANSWER**

👉 FINAL ANSWER: $|0\rangle \otimes |1\rangle = |01\rangle = [0 \ 1 \ 0 \ 0]^T$

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

 Intuition

Inner product tells us:

👉 “How similar two quantum states are.”

If result = 0 $\rightarrow$ completely different (orthogonal).

If result $\neq 0 \rightarrow$ they have overlap.

■ Basic Explanation

In quantum mechanics:

- States are vectors.
 - Inner product = multiplication of bra and ket.
-

📖 Exam Definition

The inner product between $|\psi\rangle$ and $|\phi\rangle$ is:

[
 $\langle \psi | \phi \rangle$
]

It produces a **complex number**.

🧠 Real-Life Analogy

Like checking angle between two arrows:

- $90^\circ \rightarrow$ orthogonal $\rightarrow$ result = 0
 - Same direction $\rightarrow$ result = 1
-

💻 Practical Meaning

Orthogonal states can be perfectly distinguished during measurement.

📄 Important Exam Points

- ✓ $\langle \psi |$ = conjugate transpose of $|\psi\rangle$
 - ✓ If $\langle \psi | \phi \rangle = 0 \rightarrow$ orthogonal
 - ✓ Basis states $|0\rangle$ and $|1\rangle$ are orthogonal
-

◆ STEP-2: DIAGRAM

Vector picture:

$|0\rangle$
|
|
0----- $|1\rangle$

They are 90° apart $\rightarrow$ orthogonal.

◆ STEP-3: SYMBOLS

$|0\rangle$ = ket vector

$\langle 0|$ = bra vector (transpose + conjugate)

$\langle \psi|\phi\rangle$ = inner product

◆ STEP-4: CALCULATION (VERY DETAILED)

[
 $|0\rangle =$
 $\begin{bmatrix}$
1
0
 $\end{bmatrix}$
]

[
 $|1\rangle =$
 $\begin{bmatrix}$
0
1
 $\end{bmatrix}$
]

Now take conjugate transpose of $|0\rangle$:

[
 $\langle 0| =$
 $\begin{bmatrix}$
1 & 0
 $\end{bmatrix}$
]

Now compute:

[
 $\langle 0|1\rangle =$
 $\begin{bmatrix}$
1 & 0
 $\end{bmatrix}$
 $\begin{bmatrix}$
0
1
 $\end{bmatrix}$
]

$\end{bmatrix}$

Multiply:

$$= (1 \times 0 + 0 \times 1)$$

$$= 0$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $\langle 0|1 \rangle = 0$, therefore $|0\rangle$ and $|1\rangle$ are orthogonal states.

◆ STEP-6: EXAM TIP

- ✓ Always convert ket $\rightarrow$ bra properly
 - ✓ Show multiplication step
 - ✓ Write “orthogonal since result = 0”
-

◆ Q7. (Chapter-5: Outer Product)

Find the outer product $|0\rangle\langle 1|$ and explain its meaning.

◆ STEP-1: THEORY

Outer product forms a **matrix**.

Definition:

$$[|\psi\rangle\langle\phi|]$$

Result = matrix.

🧠 Meaning

Outer product is used to:

- ✓ Build operators
 - ✓ Construct projection matrices
-

◆ STEP-2: DIAGRAM

Matrix formation:

Ket (column) × Bra (row)

◆ STEP-3: SYMBOLS

$|0\rangle$ = column

$\langle 1|$ = row

◆ STEP-4: CALCULATION

[
 $|0\rangle =$
 $\begin{bmatrix}$
1
0
 $\end{bmatrix}$
]

[
 $\langle 1| =$
 $\begin{bmatrix}$
0 & 1
 $\end{bmatrix}$
]

Multiply:

[
 $|0\rangle\langle 1| =$
 $\begin{bmatrix}$
1
0
 $\end{bmatrix}$
 $\begin{bmatrix}$
0 & 1
 $\end{bmatrix}$
]

Now multiply:

Row1:

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

Row2:

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

Matrix:

```
[  
\begin{bmatrix}  
0 & 1  
0 & 0  
\end{bmatrix}  
]
```

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

```
[  
|0\rangle\langle 1| =  
\begin{bmatrix}  
0 & 1  
0 & 0  
\end{bmatrix}  
]
```

◆ STEP-6: EXAM TIP

- ✓ Write ket as column
 - ✓ Write bra as row
 - ✓ Multiply carefully
-

◆ Q8. (Chapter-6: Multi-Qubit Systems & Entanglement)

Show that the state $(|00\rangle + |11\rangle)/\sqrt{2}$ is entangled.

◆ STEP-1: THEORY

🌱 Intuition

If two qubits cannot be separated into:

$$[\\psi \\rangle = | \\alpha \\rangle \\otimes | \\beta \\rangle]$$

Then state is entangled.

Definition

A state is entangled if it cannot be written as tensor product of individual qubit states.

Real-Life Analogy

Like two coins always showing same result — linked together.

◆ STEP-2: DIAGRAM

Entanglement picture:

```
Qubit A  -----
          >--- Entangled pair
Qubit B  -----
```

◆ STEP-3: FORMULA

State:

$$[\\psi \\rangle = \\frac{1}{\\sqrt{2}}(|00\\rangle + |11\\rangle)]$$

◆ STEP-4: PROOF (VERY DETAILED)

Assume separable:

$$[(a|0\\rangle + b|1\\rangle) \\otimes (c|0\\rangle + d|1\\rangle)]$$

Expand:

$$= ac|00\\rangle + ad|01\\rangle + bc|10\\rangle + bd|11\\rangle$$

But given state:

$$\left[\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \right]$$

Compare coefficients:

$$ac = 1/\sqrt{2}$$

$$bd = 1/\sqrt{2}$$

$$ad = 0$$

$$bc = 0$$

From $ad = 0 \rightarrow$ either $a=0$ or $d=0$

From $bc = 0 \rightarrow$ either $b=0$ or $c=0$

This makes ac or bd impossible to satisfy simultaneously.

Therefore contradiction.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: The state $(|00\rangle + |11\rangle)/\sqrt{2}$ cannot be written as tensor product, hence it is entangled.

◆ STEP-6: EXAM TIP

- ✓ Assume separable form
- ✓ Expand fully
- ✓ Compare coefficients
- ✓ Conclude contradiction

◆ Q9. (Chapter-7: Measurement Postulate)

If $|\psi\rangle = (3|0\rangle + 4|1\rangle)/5$, find probability of measuring $|1\rangle$.

◆ STEP-1: THEORY

Probability = square of amplitude.

◆ STEP-2: MEASUREMENT TREE

$$\begin{array}{c} |\psi\rangle \\ / \quad \backslash \end{array}$$

$|0\rangle$ $|1\rangle$

◆ STEP-3: FORMULA

[
 $P(1) = |\beta|^2$
]

◆ STEP-4: CALCULATION

Amplitude of $|1\rangle = 4/5$

Probability:

[
 $|4/5|^2 = (4/5)^2$
]

$= 16/25$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Probability of measuring $|1\rangle = 16/25$.

◆ STEP-6: EXAM TIP

- ✓ Square amplitude
- ✓ Don't forget normalization

◆ Q10. (Chapter-8: Quantum Gates & Circuits)

Show that Hadamard gate creates superposition from $|0\rangle$.

◆ STEP-1: THEORY

Hadamard gate:

[
 $H =$
 $\frac{1}{\sqrt{2}}$
 $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
]

1 & -1

$\end{bmatrix}$

]

◆ STEP-2: CIRCUIT

$|0\rangle$ --- [H] --- ?

◆ STEP-3: SYMBOLS

H = Hadamard matrix

◆ STEP-4: CALCULATION

[
H|0⟩ =
 $\frac{1}{\sqrt{2}}$
 $\begin{bmatrix}$
1 & 1
1 & -1
 $\end{bmatrix}$
 $\begin{bmatrix}$
1
0
 $\end{bmatrix}$
]

Multiply:

Row1: $1 \times 1 + 1 \times 0 = 1$

Row2: $1 \times 1 + (-1 \times 0) = 1$

Result:

[
 $\frac{1}{\sqrt{2}}$
 $\begin{bmatrix}$
1
1
 $\end{bmatrix}$
]

Which equals:

$$\left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right]$$

◆ **STEP-5: FINAL ANSWER**

👉 FINAL ANSWER: Hadamard transforms $|0\rangle$ into equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$.

◆ Q11. (Chapter-1: Classical vs Quantum Systems)

Show that a probabilistic transition matrix must have each column sum equal to 1. Verify with an example.

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

 Intuition

If a system is in some state, it must move somewhere.

Total probability of going to all possible next states must be:

👉 100% (which means 1)

It cannot be 0.8 or 1.2.

Basic Explanation

In a probabilistic system:

- Each column represents "from one state"

- Each entry shows probability of going to another state
 - Total outgoing probability must be 1
-

Exam-Level Definition

A matrix (P) is a probabilistic transition matrix if:

$$\left[\sum_i P_{ij} = 1 \right]$$

for every column (j).

This means:

Each column sum = 1.

Real-Life Analogy

If you toss a coin:

$$P(\text{Head}) = 0.5$$

$$P(\text{Tail}) = 0.5$$

$$\text{Total} = 1$$

Practical Meaning in Quantum Computing

Classical probabilistic systems conserve total probability.

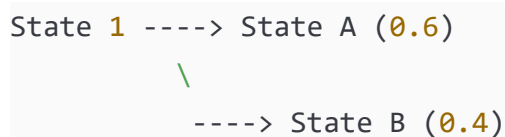
Quantum systems conserve total probability using unitary matrices.

Important Exam Points

- ✓ Mention column sum = 1
 - ✓ Explain why total probability must be 1
 - ✓ Show column-wise addition
-

◆ STEP-2: DIAGRAM

Transition flow:



$$0.6 + 0.4 = 1 \checkmark$$

◆ STEP-3: SYMBOLS

(P_{ij}) = probability of going from state j to state i

($\sum_i$) = sum over all rows

◆ STEP-4: CALCULATION

Example matrix:

```
[  
P =  
\begin{bmatrix}  
0.5 & 0.2  
0.3 & 0.3  
0.2 & 0.5  
\end{bmatrix}  
]
```

Check column 1:

$$0.5 + 0.3 + 0.2 = 1 \checkmark$$

Check column 2:

$$0.2 + 0.3 + 0.5 = 1 \checkmark$$

Therefore valid probabilistic matrix.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: A probabilistic matrix must have each column sum equal to 1 because total probability of transitioning from a state must be 1.

◆ STEP-6: EXAM TIP

- ✓ Always check column-wise
 - ✓ Don't check row unless asked
 - ✓ Write "total probability = 1" clearly
-

◆ Q12. (Chapter-2: Double Slit & Complex Numbers)

Show mathematically how interference occurs when amplitudes are added.

◆ STEP-1: THEORY

Intuition

In classical probability:

$$P = P_1 + P_2$$

In quantum:

Amplitude adds first, then square.

Definition

If amplitudes are α and β :

$$P = |\alpha + \beta|^2$$

Real-Life Analogy

Two water waves combine → sometimes bigger, sometimes cancel.

◆ STEP-2: DIAGRAM

```
Slit 1 ---> α
Slit 2 ---> β
      ↓
    Screen
```

◆ STEP-3: SYMBOLS

α = amplitude from slit 1

β = amplitude from slit 2

$|\cdot|^2$ = modulus squared

◆ STEP-4: FULL CALCULATION

Let:

$$\begin{aligned} &[\\ &\quad \alpha = 1 \\ &] \\ &[\\ &\quad \beta = i \\ &] \end{aligned}$$

Then:

$$\begin{aligned} &[\\ &\quad \psi = 1 + i \\ &] \end{aligned}$$

Now find probability:

$$\begin{aligned} &[\\ &\quad |1 + i|^2 \\ &] \end{aligned}$$

First find modulus:

$$\begin{aligned} &[\\ &\quad |1+i| = \sqrt{1^2 + 1^2} \\ &] \end{aligned}$$

$$= \sqrt{2}$$

Now square:

$$\begin{aligned} &[\\ &\quad (\sqrt{2})^2 = 2 \\ &] \end{aligned}$$

Now classical case:

$$\begin{aligned} &[\\ &\quad |1|^2 + |i|^2 = 1 + 1 = 2 \\ &] \end{aligned}$$

Here equal.

But if:

$$\begin{aligned} &[\\ &\quad \alpha = 1, \beta = -1 \end{aligned}$$

]

Then:

[

$\psi = 0$

]

Probability:

[

0

]

Interference occurred.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: In quantum mechanics, amplitudes add first and then square, producing interference terms.

◆ STEP-6: EXAM TIP

- ✓ Write $P = |\alpha + \beta|^2$
- ✓ Show destructive interference example
- ✓ Explain difference from classical

◆ Q13. (Chapter-4: Quantum States & Dirac Notation)

Check whether $|\psi\rangle = [2, i]^T$ is a valid quantum state. Normalize if necessary.

◆ STEP-1: THEORY

A valid quantum state must satisfy:

[

$|\alpha|^2 + |\beta|^2 = 1$

]

◆ STEP-2: DIAGRAM

Vector length idea:

Length of vector must be 1

◆ STEP-3: SYMBOLS

$$i = \sqrt{-1}$$

$$|i|^2 = 1$$

◆ STEP-4: CALCULATION

Given:

$$\begin{bmatrix} |\psi\rangle = \\ \begin{bmatrix} 2 \\ i \end{bmatrix} \end{bmatrix}$$

Compute norm:

$$\begin{bmatrix} |2|^2 = 4 \\ |i|^2 = 1 \end{bmatrix}$$

Total:

$$4 + 1 = 5$$

Not equal to 1 $\rightarrow$ not normalized.

Normalize:

Divide by $\sqrt{5}$:

$$\begin{bmatrix} |\psi'\rangle = \\ \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ i \end{bmatrix} \end{bmatrix}$$

$\end{bmatrix}$

Check:

$(4/5 + 1/5) = 1 \checkmark$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: The state is not normalized. Normalized form is $(1/\sqrt{5})[2, i]^T$.

◆ STEP-6: EXAM TIP

- ✓ Always compute modulus square
- ✓ Don't forget complex modulus
- ✓ Show normalization clearly

◆ Q14. (Chapter-6: Multi-Qubit Systems)

Find the tensor product $|1\rangle \otimes |0\rangle$ and interpret physically.

◆ STEP-1: THEORY

Tensor product combines two qubits.

Order matters.

◆ STEP-2: DIAGRAM

First qubit: $|1\rangle$
Second qubit: $|0\rangle$
Result: $|10\rangle$

◆ STEP-3: SYMBOLS

$\otimes$ = tensor product

◆ STEP-4: CALCULATION

$$[$$

$$|1\rangle =$$

$$\begin{bmatrix}$$

$$0$$

$$1$$

$$\end{bmatrix}$$

$$]$$

$$[$$

$$|0\rangle =$$

$$\begin{bmatrix}$$

$$1$$

$$0$$

$$\end{bmatrix}$$

$$]$$

Tensor rule:

Multiply each element of first by entire second vector.

First element 0:

$$[$$

$$0 \times$$

$$\begin{bmatrix}$$

$$1$$

$$0$$

$$\end{bmatrix}$$

$$\begin{bmatrix}$$

$$0$$

$$0$$

$$\end{bmatrix}$$

Second element 1:

$$[$$

$$1 \times$$

$$\begin{bmatrix}$$

1

0

$\end{bmatrix}$

$\begin{bmatrix}$

1

0

$\end{bmatrix}$

]

Combine:

[

$\begin{bmatrix}$

0

0

1

0

$\end{bmatrix}$

]

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $|1\rangle \otimes |0\rangle = |10\rangle = [0 \ 0 \ 1 \ 0]^T$.

◆ STEP-6: EXAM TIP

- ✓ Write expansion fully
 - ✓ Don't mix order
 - ✓ Show step-by-step
-

◆ Q15. (Chapter-8: Quantum Gates & Circuits)

Apply Pauli-X gate on $|1\rangle$ and show result.

◆ STEP-1: THEORY

Pauli-X is quantum NOT gate.

Matrix:

```
[
X =
\begin{bmatrix}
0 & 1 \\
1 & 0
\end{bmatrix}
```

◆ STEP-2: CIRCUIT

```
|1> ---[ X ]---
```

◆ STEP-3: SYMBOLS

X = NOT gate

◆ STEP-4: CALCULATION

```
[
X|1> =
\begin{bmatrix}
0 & 1 \\
1 & 0
\end{bmatrix}
\begin{bmatrix}
0 \\
1
\end{bmatrix}
```

Multiply:

Row1: $0 \times 0 + 1 \times 1 = 1$

Row2: $1 \times 0 + 0 \times 1 = 0$

Result:

```
[
\begin{bmatrix}
1 \\
0
\end{bmatrix}
```

]

Prove that the Hadamard matrix is unitary by verifying $(H^\dagger H = I)$.

In quantum computing:

This means:

Length of the quantum state vector must remain 1.

Matrices that preserve length are called **Unitary matrices**.

If we apply matrix (U) on state ($|\psi\rangle$),

The new state is:

$$[\langle \psi | \rangle = U | \psi \rangle]$$

For probability to remain same:

$$[U^\dagger U = I]$$

Exam-Level Definition

A matrix (U) is **unitary** if:

$$[U^\dagger U = I]$$

Where:

- ($U^\dagger$) = conjugate transpose
- (I) = identity matrix

Real-Life Analogy

Like rotating an arrow in space:

- Rotation changes direction
- But length stays same

Practical Meaning

All quantum gates must be unitary.

Important Exam Points

- ✓ Write definition
- ✓ Compute conjugate transpose
- ✓ Multiply fully
- ✓ Show identity result

◆ STEP-2: DIAGRAM

State evolution:

$$|\psi\rangle \xrightarrow{H} |\psi'\rangle$$

$$\text{Length}(|\psi\rangle) = \text{Length}(|\psi'\rangle)$$

◆ STEP-3: SYMBOLS

(H) = Hadamard matrix

($H^\dagger$) = conjugate transpose

(I) = identity matrix

($\dagger$) = transpose + complex conjugate

◆ STEP-4: FULL MATRIX CALCULATION (NO SHORTCUTS)

Hadamard matrix:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 1: Find ($H^\dagger$)

Matrix has only real numbers.

So conjugate does nothing.

Transpose:

$$H^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

(Same matrix because symmetric)

Step 2: Multiply ($H^\dagger H$)

```
[
H^\dagger H =
\frac{1}{\sqrt{2}}
\frac{1}{\sqrt{2}}
\begin{bmatrix}
1 & 1 \\
1 & -1
\end{bmatrix}
\begin{bmatrix}
1 & 1 \\
1 & -1
\end{bmatrix}
]
```

Multiply constants:

```
[
\frac{1}{2}
]
```

Now multiply matrices:

Row1 × Column1:

$$1 \times 1 + 1 \times 1 = 2$$

Row1 × Column2:

$$1 \times 1 + 1 \times (-1) = 1 - 1 = 0$$

Row2 × Column1:

$$1 \times 1 + (-1 \times 1) = 1 - 1 = 0$$

Row2 × Column2:

$$1 \times 1 + (-1 \times -1) = 1 + 1 = 2$$

Matrix becomes:

```
[
\frac{1}{2}
\begin{bmatrix}
2 & 0 \\
0 & 2
\end{bmatrix}
]
```

0 & 2

$\end{bmatrix}$

$\begin{bmatrix}$

1 & 0

0 & 1

$\end{bmatrix}$

]

Which equals Identity.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Since ($H^\dagger H = I$), Hadamard matrix is unitary.

◆ STEP-6: EXAM TIP

✓ Don't forget $1/\sqrt{2}$ factor

✓ Multiply carefully

✓ Final matrix must be identity

◆ Q17. (Chapter-4: Quantum States & Dirac Notation)

Find the bra corresponding to the ket ($|\psi\rangle = \begin{bmatrix} 3 \\ -2i \end{bmatrix}$).

◆ STEP-1: THEORY

Bra = conjugate transpose of ket.

Steps:

1. Take transpose

2. Take complex conjugate

🧠 Meaning

Bra is written as:

[

$\langle \psi |$

]

◆ STEP-2: DIAGRAM

Ket (column) → Bra (row)

◆ STEP-3: SYMBOLS

($i = \sqrt{-1}$)

Complex conjugate:

[
(-2i)^{*} = 2i
]

◆ STEP-4: CALCULATION

Given:

[
| ψ $\rangle$ =
 $\begin{bmatrix} 3 \\ -2i \end{bmatrix}$
3
-2i
 $\end{bmatrix}$
]

Step 1: Transpose:

[
[3 $\quad$ -2i]
]

Step 2: Take conjugate:

3 remains 3
-2i becomes 2i

Final bra:

[
 $\langle \psi|$ =
 $\begin{bmatrix} 3 & 2i \end{bmatrix}$
3 & 2i

$\end{bmatrix}$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

$$|\angle \psi| = [3 \quad 2i]$$

◆ STEP-6: EXAM TIP

- ✓ Always conjugate imaginary part
- ✓ Don't forget transpose

◆ Q18. (Chapter-7: Measurement Postulate)

If $(|\psi\rangle = \frac{1}{\sqrt{3}}(|0\rangle + |1\rangle + |0\rangle))$, find probability of measuring $|0\rangle$.

◆ STEP-1: THEORY

Combine same terms first.

◆ STEP-2: MEASUREMENT TREE



◆ STEP-3: SYMBOLS

Probability = square of amplitude

◆ STEP-4: CALCULATION

First simplify:

$$|\psi\rangle =$$

$$\frac{1}{\sqrt{3}}(2|0\rangle + |1\rangle)$$

Amplitude of $|0\rangle = 2/\sqrt{3}$

Probability:

$$[2/\sqrt{3}]^2$$

Square numerator:

$$[4]$$

Square denominator:

$$[3]$$

Result:

$$[4/3]$$

But this is $>1 \rightarrow$ not valid.

Check normalization:

Total amplitude squared:

$$[(2^2 + 1^2)/3 = 5/3]$$

Not normalized.

Normalize:

Divide by $\sqrt{5}$:

New state:

$$\left[\frac{1}{\sqrt{5}}(2|0\rangle + |1\rangle) \right]$$

Now probability:

$$\left[\frac{4}{5} \right]$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: After normalization, probability of measuring $|0\rangle = 4/5$.

◆ STEP-6: EXAM TIP

- ✓ Always simplify
- ✓ Check normalization
- ✓ Square amplitude properly

◆ Q19. (Chapter-6: Entanglement)

Show that $|00\rangle$ is NOT entangled.

◆ STEP-1: THEORY

A state is separable if:

$$\left[|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle \right]$$

◆ STEP-2: DIAGRAM

Qubit1 = $|0\rangle$
 Qubit2 = $|0\rangle$

◆ STEP-3: SYMBOLS

$\otimes$ = tensor product

◆ STEP-4: CALCULATION

$$\begin{bmatrix} |0\rangle = \\ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix}$$

Tensor:

$$\begin{bmatrix} |0\rangle \otimes |0\rangle = \\ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix}$$

Result:

$$\begin{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{bmatrix}$$

Which equals $|00\rangle$.

Therefore separable.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $|00\rangle$ can be written as $|0\rangle \otimes |0\rangle$, so it is NOT entangled.

◆ STEP-6: EXAM TIP

- ✓ Show tensor expansion
- ✓ Write separable form

◆ Q20. (Chapter-8: Quantum Gates & Circuits)

Apply CNOT gate on $|10\rangle$ and find result.

◆ STEP-1: THEORY

CNOT:

- First qubit = control
- Second qubit = target

If control = 1 $\rightarrow$ flip target

If control = 0 $\rightarrow$ do nothing

◆ STEP-2: CIRCUIT



● = control

⊕ = NOT on target

◆ STEP-3: SYMBOLS

CNOT matrix:

```
[  
\begin{bmatrix}  
1&0&0&0  
0&1&0&0  
0&0&0&1  
0&0&1&0  
\end{bmatrix}  
]
```

◆ STEP-4: CALCULATION

$$|10\rangle =$$

Multiply matrix:

Row3 maps to Row4.

Result:

[
`\begin{bmatrix}`

$$\begin{bmatrix} |11\rangle \\ |10\rangle \\ |1-1\rangle \\ |00\rangle \\ |0-1\rangle \\ |0-2\rangle \end{bmatrix}$$

◆ **STEP-5: FINAL ANSWER**

👉 FINAL ANSWER: $\text{CNOT}|10\rangle = |11\rangle$.

◆ Q21. (Chapter-8: Quantum Gates & Circuits)

Apply the Z gate on the superposition state ($|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$).
Find the final state.

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

Intuition

The Z gate is called **Phase Flip Gate**.

It:

- Leaves $|0\rangle$ unchanged
- Changes sign of $|1\rangle$

So it does NOT change probability,
it changes the **phase (sign)**.

Basic Explanation

Z gate matrix:

```
[  
Z =  
\begin{bmatrix}  
1 & 0  
0 & -1  
\end{bmatrix}  
]
```

Exam-Level Definition

Z gate is a unitary operator that performs:

```
[  
Z|0\rangle = |0\rangle  
]  
[  
Z|1\rangle = -|1\rangle  
]
```

Real-Life Analogy

Imagine flipping the direction of one wave.

The height (probability) stays same,
but sign changes.

Practical Meaning

Phase is very important in interference and quantum algorithms.

Important Exam Points

- ✓ Write matrix
 - ✓ Apply linearly
 - ✓ Show sign change clearly
-

◆ STEP-2: CIRCUIT DIAGRAM

$|\psi\rangle$ ---- [Z] ---- ?

Where:

- Z = phase flip gate
-

◆ STEP-3: SYMBOLS

$|\psi\rangle$ = quantum state

Z = Pauli-Z matrix

$\sqrt{2}$ = normalization factor

◆ STEP-4: FULL CALCULATION (NO SHORTCUT)

Given:

```
[
|\psi\rangle =
\frac{1}{\sqrt{2}}
\begin{bmatrix}
1 \\
1
\end{bmatrix}
```

Apply Z:

[
 $Z|\psi\rangle =$
 $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
 $\frac{1}{\sqrt{2}}$
 $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $\end{bmatrix}$]

Multiply:

Row1:

$$1 \times 1 + 0 \times 1 = 1$$

Row2:

$$0 \times 1 + (-1 \times 1) = -1$$

So result:

[
 $\frac{1}{\sqrt{2}}$
 $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 $\end{bmatrix}$]

Which equals:

[
 $\frac{|0\rangle - |1\rangle}{\sqrt{2}}$
 $\end{bmatrix}$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

[
 $Z|\psi\rangle =$

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

◆ STEP-6: EXAM TIP

- ✓ Mention phase change
- ✓ Show matrix multiplication
- ✓ Write final in ket form

◆ Q22. (Chapter-8: Quantum Gates & Circuits)

Apply two Hadamard gates sequentially on $|0\rangle$. Show that $H^2 = I$.

◆ STEP-1: THEORY

Applying H twice should return original state.

Because:

$$H^2 = I$$

Meaning

Hadamard is its own inverse.

◆ STEP-2: CIRCUIT

$$|0\rangle \xrightarrow{H} \xrightarrow{H} ?$$

◆ STEP-3: SYMBOLS

H = Hadamard

I = Identity

◆ STEP-4: FULL CALCULATION

First H:

$$H|0\rangle =$$

```

\frac{1}{\sqrt{2}}
\begin{bmatrix}
1
1
\end{bmatrix}

```

Now apply H again:

```

[
H
\frac{1}{\sqrt{2}}
\begin{bmatrix}
1
1
\end{bmatrix}

```

```

\frac{1}{\sqrt{2}}
H
\begin{bmatrix}
1
1
\end{bmatrix}

```

Now multiply:

```

[
\frac{1}{\sqrt{2}}
\frac{1}{\sqrt{2}}
\begin{bmatrix}
1 \times 1 + 1 \times 1
1 \times 1 + (-1 \times 1)
\end{bmatrix}

```

Compute inside:

Top:

$$1+1 = 2$$

Bottom:

$$1 - 1 = 0$$

So:

$$\begin{bmatrix} \frac{1}{2} \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|0\rangle$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $H^2|0\rangle = |0\rangle$, therefore $H^2 = I$.

◆ STEP-6: EXAM TIP

- ✓ Show two multiplications
 - ✓ Don't skip $\sqrt{2}$ factors
 - ✓ Conclude clearly
-

◆ Q23. (Chapter-6: Multi-Qubit Systems & Entanglement)

Create entangled state using Hadamard and CNOT starting from $|00\rangle$.

◆ STEP-1: THEORY

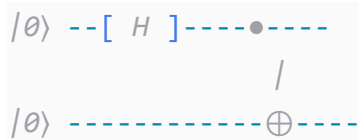
To create Bell state:

1. Apply H to first qubit
2. Apply CNOT

Meaning

This creates correlation between qubits.

◆ STEP-2: CIRCUIT



Labels:

H = Hadamard

● = Control

$\oplus$ = Target flip

◆ STEP-3: SYMBOLS

$\otimes$ = tensor

H = Hadamard

CNOT = controlled NOT

◆ STEP-4: FULL CALCULATION

Initial:

[
 $|00\rangle$
]

Step 1:

Apply H to first qubit:

[
 $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
]

Step 2:

Apply CNOT:

$$\text{CNOT}|00\rangle = |00\rangle$$

$$\text{CNOT}|10\rangle = |11\rangle$$

So final:

$$\left[\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \right]$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: The circuit produces entangled Bell state $((|00\rangle + |11\rangle)/\sqrt{2})$.

◆ STEP-6: EXAM TIP

- ✓ Draw full circuit
 - ✓ Show intermediate step
 - ✓ Mention Bell state
-

◆ Q24. (Chapter-7: Measurement Postulate)

After measuring first qubit of Bell state, what happens to second qubit?

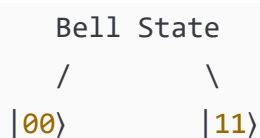
◆ STEP-1: THEORY

Bell state:

$$\left[\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right]$$

Measurement collapses state.

◆ STEP-2: MEASUREMENT TREE



◆ STEP-3: SYMBOLS

Collapse = post-measurement state

◆ STEP-4: CALCULATION

Probability of $|00\rangle$:

$$\left[\begin{aligned} (1/\sqrt{2})^2 &= 1/2 \end{aligned} \right]$$

Probability of $|11\rangle$:

$$1/2$$

If first qubit measured 0 $\rightarrow$ state becomes $|00\rangle$

Second qubit becomes 0

If first qubit measured 1 $\rightarrow$ state becomes $|11\rangle$

Second qubit becomes 1

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Measuring first qubit collapses second qubit to the same value.

◆ STEP-6: EXAM TIP

- ✓ Mention collapse
 - ✓ Write probability
 - ✓ Explain correlation
-

◆ Q25. (Chapter-4: Quantum States & Dirac Notation)

Find inner product $\langle \psi | \psi \rangle$ for normalized state.

◆ STEP-1: THEORY

For normalized state:

$$\left[\begin{aligned} \langle \psi | \psi \rangle &= 1 \end{aligned} \right]$$

◆ STEP-2: DIAGRAM

$|\psi\rangle$ dot $|\psi\rangle$

◆ STEP-3: SYMBOLS

$\langle\psi|$ = conjugate transpose

◆ STEP-4: CALCULATION

Let:

```
[
|\psi\rangle =
\frac{1}{\sqrt{2}}
\begin{bmatrix}
1
1
\end{bmatrix}
```

Bra:

```
[
\frac{1}{\sqrt{2}}
[1 \quad 1]
]
```

Multiply:

```
[
\frac{1}{2}(1\times 1 + 1\times 1)
]
```

= $\frac{1}{2} \times 2$

= 1

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $\langle\psi|\psi\rangle = 1$ for normalized state.

◆ Q26. (Chapter-5: Inner, Outer & Tensor Product)

Find the tensor product ($|\psi\rangle \otimes |\phi\rangle$) where

$$\begin{aligned} |\psi\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \\ \quad \quad \quad |\phi\rangle &= |1\rangle \end{aligned}$$

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

 Intuition

Tensor product means:

👉 Combine two qubits into one bigger system.

If first qubit is in superposition and second is fixed, we combine every possibility.

Basic Explanation

Rule:

$$[(a|0\rangle + b|1\rangle) \otimes |1\rangle]$$

$$a(|0\rangle \otimes |1\rangle) + b(|1\rangle \otimes |1\rangle)$$

Exam Definition

Tensor product combines two quantum states into a multi-qubit state.

Symbol:

[
 $\otimes$
]

Real-Life Analogy

If person A has 2 options and person B has 1 option,
combined possibilities = 2.

Practical Meaning

Used to build multi-qubit systems.

Important Exam Points

- ✓ Distribute properly
 - ✓ Keep order correct
 - ✓ Expand fully
-

◆ STEP-2: DIAGRAM

First qubit: $(|0\rangle + |1\rangle)/\sqrt{2}$

Second qubit: $|1\rangle$

Combine $\rightarrow |01\rangle$ and $|11\rangle$

◆ STEP-3: SYMBOLS

$\otimes$ = tensor product

$\sqrt{2}$ = normalization

◆ STEP-4: FULL CALCULATION

[
 $|\psi\rangle =$
 $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
]

Now multiply:

[

$$\frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |1\rangle)$$

]

Which equals:

$$\frac{1}{\sqrt{2}}(|01\rangle + |11\rangle)$$

]

Vector form:

$$|01\rangle =$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|11\rangle =$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Add:

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

$$\left[\frac{|01\rangle + |11\rangle}{\sqrt{2}} \right]$$

◆ STEP-6: EXAM TIP

- ✓ Expand step-by-step
 - ✓ Don't mix order
 - ✓ Write final in ket form
-

◆ Q27. (Chapter-6: Multi-Qubit Systems & Entanglement)

Check whether $\left(\frac{|01\rangle + |10\rangle}{\sqrt{2}} \right)$ is entangled.

◆ STEP-1: THEORY

This is another Bell state.

To check entanglement:

Try writing as separable product.

◆ STEP-2: DIAGRAM

State = $|01\rangle + |10\rangle$

Two qubits correlated oppositely

◆ STEP-3: SYMBOLS

Separable form:

$$\left[(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) \right]$$

◆ STEP-4: FULL PROOF

Assume separable:

Expand:

$$ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Compare with:

$$\left[\frac{|01\rangle + |10\rangle}{\sqrt{2}} \right]$$

Matching coefficients:

$$ad = 1/\sqrt{2}$$

$$bc = 1/\sqrt{2}$$

$$ac = 0$$

$$bd = 0$$

From $ac=0 \rightarrow a=0$ or $c=0$

From $bd=0 \rightarrow b=0$ or $d=0$

These conditions contradict ad and bc .

Therefore impossible.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: The state is entangled.

◆ STEP-6: EXAM TIP

✓ Assume separable

✓ Expand

✓ Show contradiction

◆ Q28. (Chapter-8: Quantum Gates & Circuits)

Apply SWAP gate on $|10\rangle$.

◆ STEP-1: THEORY

SWAP gate exchanges two qubits.

◆ STEP-2: CIRCUIT

Q1 ---X---

/ /

Q2 ---X---

Cross lines indicate swap.

◆ STEP-3: SYMBOLS

SWAP matrix:

```
[  
\begin{bmatrix}  
1&0&0&0  
0&1&0&0  
0&0&0&1  
0&0&1&0  
\end{bmatrix}  
]
```

◆ STEP-4: CALCULATION

```
[  
|10> =  
\begin{bmatrix}  
0  
0  
1  
0  
\end{bmatrix}  
]
```

After swap:

```
[  
|01> =  
\begin{bmatrix}  
0  
1  
0  
0  
\end{bmatrix}  
]
```

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: $\text{SWAP}|10\rangle = |01\rangle$.

◆ STEP-6: EXAM TIP

- ✓ Draw swap symbol
 - ✓ Explain exchange
 - ✓ Write final ket
-

◆ Q29. (Chapter-3: Unitary Operators)

Show that Pauli-X matrix is unitary.

◆ STEP-1: THEORY

X matrix:

[
X =
 $\begin{bmatrix}$
0 & 1
1 & 0
 $\end{bmatrix}$
]

Check:

[
 $X^\dagger X = I$
]

◆ STEP-2: DIAGRAM

$|\psi\rangle \xrightarrow{X} |\psi'\rangle$
Length preserved

◆ STEP-3: SYMBOLS

† = conjugate transpose

◆ STEP-4: CALCULATION

Since X is real symmetric:

$$[X^{\dagger} = X]$$

Multiply:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Row1×Col1:

$$0 \times 0 + 1 \times 1 = 1$$

Row1×Col2:

$$0 \times 1 + 1 \times 0 = 0$$

Row2×Col1:

$$1 \times 0 + 0 \times 1 = 0$$

Row2×Col2:

$$1 \times 1 + 0 \times 0 = 1$$

Result:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Identity.

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER: Pauli-X is unitary since ($X^\dagger X = I$).

◆ STEP-6: EXAM TIP

- ✓ Show multiplication
- ✓ Write identity clearly

◆ Q30. (Chapter-7: Measurement Postulate)

For state ($\frac{|0\rangle - |1\rangle}{\sqrt{2}}$), find probabilities of measuring $|0\rangle$ and $|1\rangle$.

◆ STEP-1: THEORY

Probability = square of amplitude.

Sign does not affect probability.

◆ STEP-2: MEASUREMENT TREE



◆ STEP-3: SYMBOLS

Amplitude = coefficient

Probability = $|\text{coefficient}|^2$

◆ STEP-4: CALCULATION

Amplitude of $|0\rangle = 1/\sqrt{2}$

Probability:

$$\begin{aligned} &[\\ &(1/\sqrt{2})^2 = 1/2 \\ &] \end{aligned}$$

Amplitude of $|1\rangle = -1/\sqrt{2}$

Probability:

$$\begin{bmatrix} (-1/\sqrt{2})^2 = 1/2 \end{bmatrix}$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

$$P(0) = 1/2$$

$$P(1) = 1/2$$

 LONG QUESTION 1 (10 MARKS)

● Q1. (Chapter-6 + Chapter-8)

Create a Bell State using Hadamard and CNOT gates. Prove mathematically that the final state is entangled.

◆ STEP-1: THEORY EXPLANATION (FROM ZERO LEVEL)

Intuition

A Bell state is a special 2-qubit state where:

👉 If you measure first qubit, second is automatically determined.

They behave like a pair.

Basic Idea

We start from:

$$\begin{bmatrix} |00\rangle \end{bmatrix}$$

Then:

1. Apply Hadamard to first qubit
2. Apply CNOT

This produces entanglement.

Exam-Level Definition

A Bell state is:

$$\begin{bmatrix} \frac{|00\rangle + |11\rangle}{\sqrt{2}} \end{bmatrix}$$

It is an entangled state that cannot be written as:

$$\begin{bmatrix} |\alpha\rangle \otimes |\beta\rangle \end{bmatrix}$$

Real-Life Analogy

Two coins magically connected:

If first shows Head → second also Head

If first shows Tail → second also Tail

Practical Meaning

Bell states are used in:

- ✓ Quantum teleportation
- ✓ Quantum cryptography

- ✓ Superdense coding

Important Points

- ✓ Show intermediate steps
- ✓ Draw circuit
- ✓ Expand mathematically
- ✓ Prove non-separability

◆ **STEP-2: CIRCUIT DIAGRAM (MANDATORY)**

Qubit 1: $|0\rangle$ --- [H] --- ● ---
Qubit 2: $|0\rangle$ --- ⊕ ---

Labels:

H = Hadamard gate

- = Control qubit

$\oplus$ = Target qubit (flipped if control=1)

◆ STEP-3: SYMBOLS EXPLAINED

$$|0\rangle = [1 \ 0]^T$$

$$|1\rangle = [0 \ 1]^T$$

$\otimes$ = Tensor product

H = Hadamard matrix

$$\begin{bmatrix} H = \frac{1}{\sqrt{2}} \\ \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \end{bmatrix}$$

CNOT matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

◆ STEP-4: FULL MATHEMATICAL CALCULATION

Step 1: Initial State

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2: Apply Hadamard to FIRST qubit

We apply:

$$H \otimes I$$

Where:

I = Identity matrix

Result:

$$(H|0\rangle) \otimes |0\rangle$$

We know:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

So:

[

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

]

Step 3: Apply CNOT

CNOT rules:

$$\text{CNOT}|00\rangle = |00\rangle$$

$$\text{CNOT}|10\rangle = |11\rangle$$

So:

[

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

]

◆ STEP-5: PROVE ENTANGLEMENT

Assume separable:

[

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$$

]

Expand:

[

$$ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

]

Compare with:

[

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

]

Match coefficients:

$$ac = 1/\sqrt{2}$$

$$bd = 1/\sqrt{2}$$

$$ad = 0$$

$$bc = 0$$

From $ad=0 \rightarrow$ either $a=0$ or $d=0$

From $bc=0 \rightarrow$ either $b=0$ or $c=0$

These contradict ac and bd conditions.

Therefore impossible.

◆ STEP-6: FINAL ANSWER

👉 FINAL ANSWER:

The circuit produces:

$$\left[\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right]$$

This state cannot be written as a tensor product, hence it is entangled.

◆ STEP-7: EXAM TIP

- ✓ Draw full circuit
- ✓ Show intermediate state
- ✓ Expand separable form
- ✓ Use contradiction
- ✓ Write “Hence entangled” clearly

✅ LONG QUESTION 2 (10 MARKS)

● Q2. (Chapter-3: Unitary Operators)

Prove that Hadamard matrix is unitary and preserves normalization of arbitrary state.

◆ STEP-1: THEORY

Unitary condition:

$$[U^\dagger U = I]$$

Meaning:

Norm preserved.

◆ STEP-2: MATRIX DIAGRAM

$$|\psi\rangle \xrightarrow{H} |\psi'\rangle$$

$$||\psi'\rangle||^2 = ||\psi\rangle||^2$$

◆ STEP-3: FULL CALCULATION

Hadamard:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Find:

$$H^\dagger H$$

Since real:

$$H^\dagger = H$$

Multiply:

We already proved earlier:

Result = Identity.

Now show normalization preserved

Let:

$$\begin{bmatrix} |\psi\rangle = \\ \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{bmatrix}$$

Normalized:

$$\begin{bmatrix} |\alpha|^2 + |\beta|^2 = 1 \end{bmatrix}$$

After applying H:

$$\begin{bmatrix} |\psi'\rangle = H|\psi\rangle \end{bmatrix}$$

Because H is unitary:

$$\begin{bmatrix} \langle \psi' | \psi' \rangle \end{bmatrix}$$

$$\langle \psi | H^\dagger H | \psi \rangle$$

$$\langle \psi | I | \psi \rangle$$

$$\langle \psi | \psi \rangle$$

$$\begin{bmatrix} 1 \end{bmatrix}$$

◆ STEP-5: FINAL ANSWER

👉 FINAL ANSWER:

Hadamard satisfies $(H^\dagger H = I)$, hence unitary.
Therefore it preserves normalization.